Response to the Editor and Reviewers

Joint Evidence Gating for Continual Maintenance of Predictive Digital Twin Models Under Unreliable Monitoring

Computers, Materials & Continua

Dear Editor and Reviewers,

Thank you for your comments. Below we answer each point, summarize the relevant evidence and identify its location in the revised Clean manuscript. Supplement references refer to the accompanying Supplementary Protocol and Results.

Response to Reviewer 1

R1.1

Comment. Section 3 frames RAVM-DT as a general algorithm for "a DT platform that maintains center-specific models for a set of heterogeneous computing centers." However, all experimental evaluation — including the Alibaba Cluster Trace in Section 5.7 — addresses the same task: workload (resource utilization) prediction for computing centers. The generality claim is therefore unsupported by the evidence. Digital twin maintenance problems arise in power grids, manufacturing processes, autonomous vehicles, and other domains with qualitatively different data characteristics, drift patterns, and corruption modes. Without at least one additional DT task domain, the reader cannot determine whether RAVM-DT's safety–utility trade-off is domain-specific or genuinely general.

Response. The study concerns predictive DT model maintenance, with workload and measured temperature forecasting as distinct task families. The added thermal evaluation uses three UCI room targets from one home and two ThermBuild houses at one site. Every tested thermal stream fails the transferred source risk criterion. On the 40 previously exposed Alibaba machines, Joint meets the empirical risk criterion but loses the relative-utility comparison with DualMean. These results support the narrower predictive-maintenance scope and are retained in the abstract, results and discussion.

Changes in the revised manuscript. Section 1 (p. 1); Section 5.1.1 (p. 6); Section 5.2.6 (p. 13); Table 7 (p. 14); Section 5.3 (p. 15).

R1.2

Comment. Figure 3(d) shows that the Validation-Commit (Val.-Commit) baseline achieves a 0.0% commit rate across all five primary conditions, it never deploys any candidate throughout the evaluation. As a result, Val.-Commit is functionally identical to Frozen in MSE and harmful transitions (both at 4.843×10^{-3} and 0.0% harmful transitions, respectively).

Response. The original ValidationCommit baseline uses a calibrated retain-all fallback and consequently commits zero candidates. None of its five tested thresholds met the mixed-stream harm criterion; even the largest yielded 123 harmful commitments in 1,440 opportunities (8.54%). At threshold zero, the original clean/mixed calibration committed 869/1,440 and 1,054/1,440 candidates, including 142 and 363 harmful commits. Supplement Table S2 retains that evidence separately from Main. The revised comparison includes active RecentMean, MWU-Mean and DualMean controls, and reports the candidate-generation cost of inactive ValidationCommit.

Changes in the revised manuscript. Section 5.1.2 (p. 7); Table 2 (p. 9); Section 5.1.3 (p. 8); Table 8 (p. 15). Supplement S2, Table S2.

Response to Reviewer 2

R2.1

Comment. The novelty relative to existing validation/gated update approaches must be clearly established.

Response. Joint Gate combines the magnitude of current mean improvement with scaled rank evidence, subject to a separate negative-mean condition. A sufficiently large mean improvement can offset weaker rank evidence; this differs from MWU-Mean's hard conjunction and DualMean's historical condition. On 6,912 identical states, Joint differs from DualMean in 971 actions (731 Joint-only, 240 Dual-only). Main shows the stated risk-qualified utility benefit; replay establishes a behavioral distinction, not an independent causal contribution for each term.

The literature comparison distinguishes decision objects: ADWIN revises a temporal data window, whereas Seldonian regression applies a held-out constraint test under its sampling assumptions. RecentMean, MWU-Mean and DualMean are explicitly study-defined controls, not reproductions of those published algorithms. The complete Joint formula and implementation are in the Methods section.

Changes in the revised manuscript. Section 2 (p. 2); Section 4 (p. 4); Equation 4 (p. 4); Section 4.2 (p. 5); Section 5.2.3 (p. 10); Section 5.3 (p. 15). Supplement S7.

R2.2

Comment. The Comparison against ValidationCommit deserves further study.

Response. ValidationCommit's zero-commit outcome follows its disclosed calibration fallback. Supplement S2 reports all five original thresholds and streams; Table 2 compares all twelve Main methods, including active mean-based alternatives and inactive Frozen/ValidationCommit. The principal active comparator is DualMean, selected among empirically qualified active alternatives. That selection and its uncertainty limitation are stated in the calibration subsection.

Changes in the revised manuscript. Section 5.1.2 (p. 7); Section 5.1.3 (p. 8); Table 2 (p. 9); Figure 6 (p. 12). Supplement S2.

R2.3

Comment. The calibration process must be made more transparent.

Response. The protocol specifies source pretraining, shared target warm-up, candidate adaptation, current Verification and later Service evaluation. Main gives the data roles, drift regimes, model structures, SGD settings and intervention rules; Supplement S1–S3 supplies the full recipes. Joint uses a sixteen-cell grid: $\lambda \in \{0, .05, .15, .3\}$ and $\delta \in \{1, 5, 10, 20\}\eta$. Eleven cells meet risk eligibility and nine enter the active positive-weight selection pool. The coherent-stream harm cap is 4.5%; selection then minimizes balanced MSE, with the stated cost/weight/margin/identifier tie-breaks. Neighboring gates have separate margin searches. Supplement S6 distinguishes the stopped/corrected development history from Main evaluation.

Changes in the revised manuscript. Section 5.1.1 (p. 6); Section 5.1.2 (p. 7); Section 5.1.3 (p. 8); Equation 4 (p. 4); Section 5.2.5 (p. 12). Supplement S1–S3.

R2.4

Comment. The corruption model may not fully capture real monitoring failures. The authors should provide justification for these corruption models and the parameter ranges based on actual digital twin monitoring systems.

Response. The interventions are now identified as controlled stress settings. Published sensor studies support noise, abnormal constant readings, missingness and delay; HVAC benchmarks include simulated $\pm 2/\pm 4\,^{\circ}\mathrm{C}$ biases. Under source-only scaling, a 2°C offset corresponds to approximately 2.13136/1.83461 standardized units in BSE1/BSE2. Supplement S9 provides the parameter-to-evidence ledger, native-quality profiles and numerical sources.

The additional public field evidence includes three AHU records and QUANT paired-reference measurements (591,316 finite NO_2 pairs from 55 systems in three cities). AHU operational labels lack paired calibration truth; QUANT discrepancies use different physical units and tasks. These records provide numerical context but do not establish the target DT incidence, amplitude or duration. The 5/10/20% fractions, .2 offsets, replay and one-hour episodes remain stress-test choices; full target-site numerical calibration is unresolved. The associated descriptive scripts are supplied without changing the experiments.

Changes in the revised manuscript. Section 5.1.5 (p. 8); Section 5.2.2 (p. 9); Table 3 (p. 10); Figure 4 (p. 11); Section 5.3 (p. 15). Supplement S3, S7, S9; numerical evidence and raw-quality CSVs.

R2.5

Comment. The assumptions about clean service data deserve more discussion.

Response. Service supplies retrospective reference outcomes outside the injected monitoring fault. The action and model identities are sealed before Service access; matured data may later enter legal Update roles. For measured data, Service is the observed record, and missing thermal targets remain unscored. The discussion explains how delayed, biased or missing reference labels can affect risk estimates and maintenance feedback despite correct temporal access control.

Changes in the revised manuscript. Section 3 (p. 3); Figure 2 (p. 5); Section 3.2 (p. 4); Section 5.3 (p. 15).

R2.6

Comment. The safety claim must be carefully qualified.

Response. We use the term empirical risk qualification. Harmful-commit frequency and mean downside divide by all maintenance opportunities, including retained candidates. The source criteria are $h \leq .05$ and $D \leq \eta$, with $\eta = 5.6036475532487204 \times 10^{-5}$, separately in each required stream. They are observed evaluation conditions rather than future-risk guarantees. The rank output is likewise a score feature under dependent windows. Results retain qualification failures under severe Mixed corruption, all three H4 Mixed groups and every tested thermal stream; Alibaba qualifies but loses the relative-utility comparison.

Changes in the revised manuscript. Section 3.2 (p. 4); Equation 2 (p. 4); Section 4 (p. 4); Section 5.2.2 (p. 9); Section 5.2.4 (p. 12); Section 5.2.6 (p. 13); Section 5.3 (p. 15).

R2.7

Comment. The external-domain performance raises a significant limitation, which merits further discussion.

Response. On 40 previously exposed Alibaba machines, Joint meets the transferred risk criterion in clean and mixed streams: harmful commits are 16/2,880 (0.556%) and 114/2,880 (3.958%), with downside 3.18493×10^{-7} and 7.66545×10^{-6}. DualMean also qualifies and has lower MSE, harmful frequency and downside. Joint-minus-DualMean balanced MSE is $+1.26433 \times 10^{-5}$ with unadjusted 95% machine-paired interval $[3.88690 \times 10^{-6}, 2.20413 \times 10^{-5}]$. Thus source risk qualification transfers to these streams, but the relative-utility advantage does not. Prior exposure limits independent confirmation. The thermal evaluation separately fails transferred risk qualification in every tested stream.

Changes in the revised manuscript. Section 5.2.6 (p. 13); Table 6 (p. 13); Table 2 (p. 9); Section 5.2.7 (p. 14); Table 8 (p. 15); Section 5.3 (p. 15). Supplement Table S5.

R2.8

Comment. The trade-off between utility and safety must be clearly stated.

Response. Main Joint reduces balanced MSE by 4.85% relative to the selected empirically qualified active DualMean comparator, improving in 29/32 centers while committing at 19.0538% of clean/mixed opportunities. Always-FT, EWC, ER and Huber-FT have lower raw MSE but fail the source risk criterion. The RecentMean contrast crosses zero, so the evidence does not establish uniform utility superiority. The results also report missed beneficial candidates and method-specific recall denominators. The discussion explains this application-dependent trade-off; the separate cost evaluation finds Joint slower than DualMean (14.28 versus 11.70 s per 24-stage lifecycle).

Changes in the revised manuscript. Section 5.2.1 (p. 9); Table 2 (p. 9); Section 5.2.3 (p. 10); Figure 5 (p. 11); Section 5.2.7 (p. 14); Table 8 (p. 15); Section 5.3 (p. 15). Supplement S6.

R2.9

Comment. More details need to be provided for reproduction.

Response. The attachment supplies read-only aggregate reconstruction and an executable synthetic subset: five group configurations, required processed inputs, source/M0 checkpoints, original code, environment locks and relative-path commands. It supports sixteen previously accepted

synthetic PATH_A lifecycles and one source-fitting-to-lifecycle PATH_B. Supplement S8–S10 and `training_bundle/REPRODUCE.md` give commands, stage-specific environments and exact scope.

Historical PATH_A acceptance covered nineteen lifecycles/472 stages, including three measured-domain cases whose input/checkpoint bytes are excluded from the submitted subset. All original audits passed, with matching model states and actions and maximum absolute MSE difference zero. The specified synthetic PATH_B regenerated the source and warm-up states, then completed one 24-stage Joint lifecycle; source/M0 states, 72 lifecycle state comparisons and all 24 actions matched, with maximum MSE difference zero. These are engineering checks of existing cases. Full matrices, other-domain source paths, development selection and independent external reproduction remain unvalidated. Provider acquisition instructions and rights apply to excluded third-party assets.

Changes in the revised manuscript. Section 5.1.1 (p. 6); Section 5.1.2 (p. 7); Section 5.1.3 (p. 8); Section 5.1.4 (p. 8); Section 5.2.7 (p. 14). Supplement S1–S10; attached 08_Reproduction/README.md and training_bundle/REPRODUCE.md.

R2.10

Comment. The scope of the conclusions must be kept appropriately narrow.

Response. The title, problem formulation and contribution statements focus on predictive-model maintenance using current Verification evidence. Main supports a risk-qualified utility benefit in the synthetic workload setting. Monitoring, horizon, external-workload and thermal results specify its limits. The discussion consolidates the implications for domain-specific development and validation, and the conclusion uses the same scope.

Changes in the revised manuscript. Section 1 (p. 1); Section 3 (p. 3); Section 5.3 (p. 15); Section 6 (p. 16).

R2.11

Comment. The description of parameter sensitivity needs to be improved.

Response. Figure 7(a,b) shows all sixteen frozen development configurations, including ineligible cells. Eleven meet risk eligibility; nine active positive-weight cells enter Joint selection. The selected rule uses $\lambda = .3$ and $\delta = 5.6036475532487204 \times 10^{-5}$. The text explains how margin and rank weight affect activity and qualification. Figure 7(c), now explicitly cited in the architecture/horizon results, shows all six E2 contrasts, including the three H4 risk failures. Selection precedes these extended evaluations.

Changes in the revised manuscript. Section 5.1.3 (p. 8); Section 5.2.5 (p. 12); Figure 7 (p. 13).

R2.12

Comment. Some minor presentation issues exist. The figures are quite clear, especially the workflow in Figure 2 and the mechanism/sensitivity analyses. However, many plots use fairly small labels. It will help to increase label sizes and make legend descriptions simpler. Notation should also be carefully checked.

Response. Figures 1–8 and all tables were checked at their final manuscript sizes. Figure 1 uses the original submission's scale of 88% of the text width. Labels in the experimental plots remain at least 8pt. Captions retain panel meanings, denominators and interval encoding without repeating the result discussion. Figure 2 is editable vector artwork: both models enter common current-Verification loss

evaluation, both evidence outputs enter the Joint Score, and Service auditing follows sealing. SVG, vector PDF and publication TIFF are supplied. Score notation, figure references and the corrected Alibaba citation were checked against the compiled Clean.

Changes in the revised manuscript. Figure 2 (p. 5); Figure 6 (p. 12); Figure 4 (p. 11); Figure 7 (p. 13); Section 3.2 (p. 4); Section 4 (p. 4). Supplement S10; editable Figure 2 artwork.

Response to Editor 1

E1.1

Comment. Figure 5 is not cited in the main text. Please add an appropriate in-text citation to Figure 5 at the relevant location.

Response. The text explicitly cites the current benign-evolution Figure 5 and Table 4. Figure 5(a) compares clean lifecycle MSE for Always-FT, Joint and Frozen; panel(b) reports Joint commitment and beneficial-candidate recall, using method-specific candidate-chain denominators.

Changes in the revised manuscript. Section 5.2.3 (p. 10); Table 4 (p. 11); Figure 5 (p. 11).

E1.2

Comment. Section 4.1.1: Please verify the accuracy and completeness of the information provided in Reference [32] and make corrections if necessary.

Response. The corrected entry identifies the Alibaba Open Cluster Trace Program, Cluster Trace v2018, and its official documentation. The dataset description distinguishes the full provider trace from the 40 previously exposed machines evaluated here.

Changes in the revised manuscript. Section 5.1.1 (p. 7). Corrected Alibaba dataset entry in the Clean References.